# PASS SCIENCE WORKS - ASSIGNMENT

## 1. Problem Statement

Consider a single-sample regression problem using a neural network with one input neuron, one hidden neuron and one output neuron.

Input $x = 1.0$
Target $y = 2.0$
Input weight $w = 0.5$
Hidden bias $b = 0.0$
Output weight $v = 0.5$
Output bias $c = 0.0$
Learning rate $\eta = 0.1$

Objective: minimize the prediction error by updating the network parameters using backpropagation & gradient descent.

Foward Propagation: ① Hidden layer input

$$z = wx + b$$
$$z = (0.5 \times 1) + 0 = 0.5$$

② Applying tanh function

$$h = \tanh(z)$$
$$h = \tanh(0.5) = 0.462117$$

③ Calculating predicted output

$$\hat{y} = vh + c$$
$$\hat{y} = (1 \times 0.462117) + 0$$
$$\boxed{\hat{y} = 0.462117}$$

④ Calculating loss

$$J = \frac{1}{2}(\hat{y} - y)^2$$
$$J = \frac{1}{2}(0.462117 - 2)^2$$
$$\boxed{J = 1.182542}$$

The predicted output is 0.462117, but the target is 2.0. Therefore, the network needs to adjust its weights & biases to reduce the loss.

## Backpropagation (Chain Rule)

$$J = \frac{1}{2}(\hat{y} - y)^2$$

Differentiating, $\frac{\partial J}{\partial \hat{y}} = \hat{y} - y$

$$= 0.462117 - 2 = 1.537883$$

a) Gradient of output weight (V)

$$\frac{\partial J}{\partial V} = \frac{\partial J}{\partial \hat{y}} \times h \Rightarrow (-1.537883)(0.462117)$$

$$\boxed{\frac{\partial J}{\partial V} = -0.710682}$$

b) Gradient of output bias (c)

$$\frac{\partial J}{\partial c} = \frac{\partial J}{\partial \hat{y}} \times 1 \quad \left| \frac{\partial J}{\partial z} = 1 - \tanh^2(z) \right.$$

$$= 1 - (0.42117)^2 = 0.786448$$

$$\therefore (-1.537883)(1)(0.786448)$$

$$= -1.209464$$

Input gradient weight

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial z} \times x$$

$$\boxed{\frac{\partial J}{\partial w} = -1.209464}$$

hidden bias gradient $\frac{\partial J}{\partial b} = \frac{\partial J}{\partial z} \times 1$

$$\boxed{\frac{\partial J}{\partial b} = -1.209464}$$

The Update Formula is $\theta_{new} = \theta_{old} - \eta \frac{\partial J}{\partial \theta}$

Learning rate $\eta = 0.1$

$W_{new} = 0.5 - (0.1 \times -1.209464)$

$W_{new} = 0.620946$

$b_{new} = 0 - (0.1 \times -1.209464)$

$b_{new} = 0.120946$

$V_{new} = 1 - (0.1 \times 0.710682)$

$V_{new} = 1.071068$

$C_{new} = 0 - (0.1 \times -1.537883)$

$C_{new} = 0.153788$

Updated parameters:

$W = 0.620946$

$b = 0.120946$

$V = 1.071068$

$C = 0.153788.$

Backpropagation calculates gradients using the chain rule, & gradient descent uses these gradients to update the network parameters & reduce prediction loss. 2